A tool for learning UML with a tutor that gives hints without revealing the solution

Questionnaire for modeling instructors · about 20 minutes

Thank you for taking part. We developed **ModelaU**, a workspace for learning UML with a tutor based on a large language model (LLM). First, you will watch a four-minute video demonstrating the tool with a new class diagram exercise. Then we will ask for your opinion as a modeling instructor. There are no right or wrong answers.

Your responses are anonymous and will be analyzed in aggregate. We do not ask for your name or contact details, and the server does not store your IP address. You can leave at any time. You must answer the profile and rating questions to continue. For rating questions, you can select "I don't know / not shown in the video". Written comments are optional. By submitting, you agree to the use of your responses in aggregate in a scientific publication.

Demo video

There is no narration. On-screen text explains each scene. Please watch the full video before answering.





The problem · 0:00 | Our proposal · 0:24 | The eight mechanisms · 0:53 | Demonstration · 1:30 | Summary · 3:40

Video text: Español | English

Would you like to try ModelaU?

If you would like to explore the tool before answering, open modelau.duckdns.org ↗. It opens in a new tab so you can return to the questionnaire. Trying the tool is optional.

To use the tutor with the shared model, without supplying your own API key:

1. Open **LLM settings**.
2. For the provider, select **Server default**.
3. Enter the password `coffee` and click **Unlock**.
4. Save your settings.
5. In the **Exercise** panel, click **Generate** to create an exercise prompt. You can enter a topic or leave it blank for a random one. For a first try, select **Simple** and click **Generate**.
6. Start building the diagram by adding a few classes and relationships. Then ask **UMLBuddy** for a hint and see how it uses your exercise prompt and diagram to help.

When you are finished, return to this tab and continue the questionnaire.

I watched the video · Continue →

About you

Answer the profile questions to continue.

How many years have you been teaching UML or modeling?

- ○ 0 to 2
- ○ 3 to 5
- ○ 6 to 10
- ○ More than 10

In which courses do you teach or have you taught UML or modeling?

At which university do you teach or have you taught UML or modeling?

Have you used AI chat assistants in your teaching?

- ○ Never

- ○ Occasionally
- ○ Regularly

Did you watch the entire video?

- ○ All of it
- ○ Most of it
- ○ Only part of it

How have you used ModelaU?

- ○ I haven't used it
- ○ I explored it without working on an exercise
- ○ I used it to work on one exercise
- ○ I used it to work on several exercises

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▶ Watch the video

ModelaU compared with conventional chat

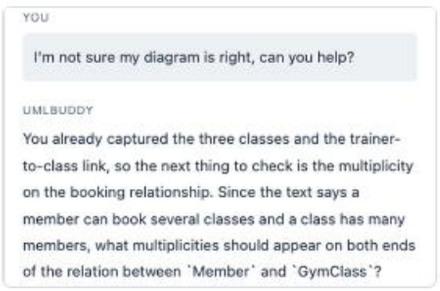
Consider the feedback the student receives in each case. Rate each statement from 1 to 5 for ModelaU and for the conventional chat in the video. Both responses remain visible as you answer.

Conventional chat (separate from ModelaU)



▶ Watch in the video (0:06)

ModelaU



▶ Watch in the video (2:27)

The four qualities in Nicol's theory

Nicol (2010) describes effective feedback in terms of these qualities. Rate how well each is met.

Specific The feedback points to the exact part of the student's work it addresses (for example, a particular class, relationship, or step of the diagram), rather than giving general comments such as "review your diagram".

CONVENTIONAL CHAT	MODELAU
1 2 3 4 5	1 2 3 4 5
I don't know / not shown in the video	I don't know / not shown in the video

Selective The feedback focuses on a single issue per interaction, the most important at that moment, rather than listing every issue that could be raised at once.

CONVENTIONAL CHAT	MODELAU
1 2 3 4 5	1 2 3 4 5
I don't know / not shown in the video	I don't know / not shown in the video

Timely The feedback arrives while the student's attempt is still in progress (for example, when they seem stuck or make a structural mistake), and not only when they explicitly ask, which may be too late.

CONVENTIONAL CHAT	MODELAU
1 2 3 4 5	1 2 3 4 5
I don't know / not shown in the video	I don't know / not shown in the video

Contextualized The feedback adapts to the stage of the task the student is at (for example, identifying classes or defining relationships), without imposing a fixed order between stages.

CONVENTIONAL CHAT	MODELAU
1 2 3 4 5	1 2 3 4 5
I don't know / not shown in the video	I don't know / not shown in the video

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▶ Watch the video

The eight mechanisms

Each mechanism targets one of Nicol's four qualities of effective feedback. For each one, rate from 1 to 5 whether it achieves that quality and whether it would get in the student's way. The image on the left shows what it looked like. Use the button to replay that moment.

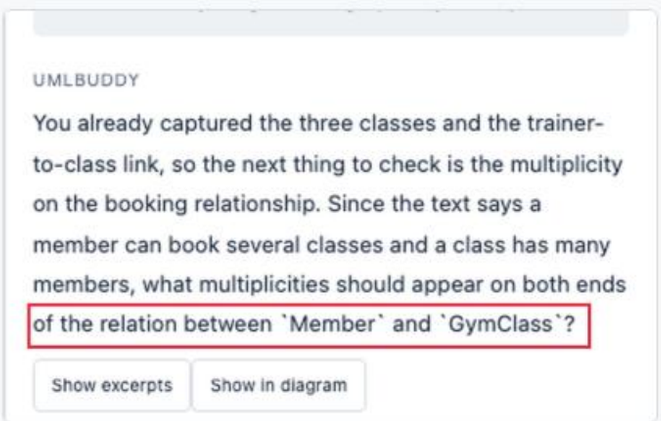
The four qualities at a glance

Specific
Points to the exact part of the student's work it addresses (for example, a particular class, relationship, or step of the diagram), rather than giving general comments such as "review your diagram".

Selective
Focuses on a single issue per interaction, the most important at that moment, rather than listing every issue that could be raised at once.

Timely
Arrives while the student's attempt is still in progress (for example, when they seem stuck or make a structural mistake), and not only when they explicitly ask, which may be too late.

Contextualized
Adapts to the stage of the task the student is at (for example, identifying classes or defining relationships), without imposing a fixed order between stages.



In the video, the response refers to "Member" and "GymClass" by their names in the diagram.

▶ Watch this moment in the video (2:37)

Specific: the feedback points to the exact part of the student's work it addresses (for example, a particular class, relationship, or step of the diagram), rather than giving general comments such as "review your diagram". This tells the student where to look and what to revise, instead of leaving them to guess.

1. Referencing exact names

Specific

The tutor refers to the elements of the student's work (classes, attributes, actors, or use cases) by the exact names they have in the diagram, rather than vague phrases such as "that class" or "that relationship". The aim is for the student to know without doubt which part of their work is being discussed.

This mechanism helps feedback be specific.

1 2 3 4 5

I don't know / not shown in the video

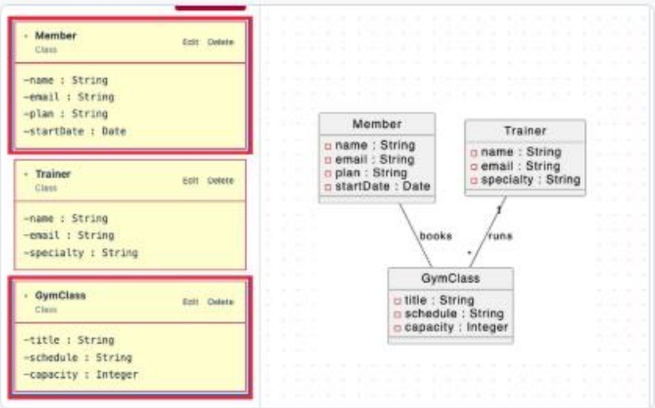
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



Clicking "Show in diagram" outlines the elements the tutor mentions in blue. Highlighting is limited to elements found in the diagram.

▶ Watch this moment in the video (2:49)

Specific: the feedback points to the exact part of the student's work it addresses (for example, a particular class, relationship, or step of the diagram), rather than giving general comments such as "review your diagram". This tells the student where to look and what to revise, instead of leaving them to guess.

2. Highlighting verified diagram elements

Specific

When the tutor mentions diagram elements, the interface outlines them so the student can find them at a glance. Before outlining them, the server checks that each one really exists in the diagram, so nothing invented by the model is ever pointed at.

This mechanism helps feedback be specific.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

YOU

I'm not sure my diagram is right, can you help?

UMLBUDDY

You already captured the three classes and the trainer-to-class link, so the next thing to check is the multiplicity on the booking relationship. Since the text says a member can book several classes and a class has many members, what multiplicities should appear on both ends of the relation between `Member` and `GymClass`?

Show excerpts Show in diagram

The response is brief, focuses on one issue, and ends with a question so the student can work out what to add.

▶ Watch this moment in the video (2:27)

Selective: the feedback focuses on a single issue per interaction, the most important at that moment, rather than listing every issue that could be raised at once. This lets the student act on each comment without being overwhelmed by more than they can handle at once.

3. One issue per response

Selective

Each response focuses on a single issue, the most relevant at that moment, and ends with a question. Other possible issues wait for later interactions. The aim is for the student to act on the comment without being overwhelmed by a list of corrections.

This mechanism helps feedback be selective.

1 2 3 4 5

I don't know / not shown in the video

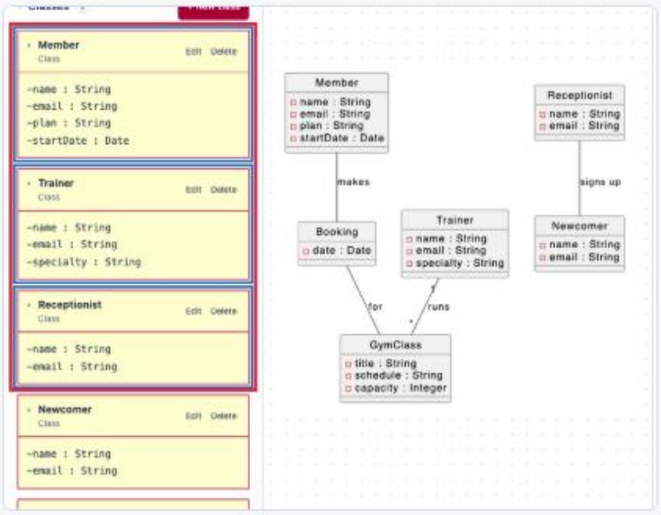
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



With a more advanced diagram, the student asks the tutor to show all the classes that repeat data. The model proposed four (Newcomer too), but only Member, Trainer, and Receptionist are outlined; the fourth is left for the student to find.

▶ Watch this moment in the video (3:24)

Selective: the feedback focuses on a single issue per interaction, the most important at that moment, rather than listing every issue that could be raised at once. This lets the student act on each comment without being overwhelmed by more than they can handle at once.

4. At most three highlights per response

Selective

The interface outlines at most three diagram elements per response, even if the problem affects more. The marks serve as examples: the student sees a few and can be the one to find the others, instead of having everything pointed out. The aim is not to give the student more than they can handle at once; the tutor's message may name more elements, but only three are marked.

This mechanism helps feedback be selective.

1 2 3 4 5

I don't know / not shown in the video

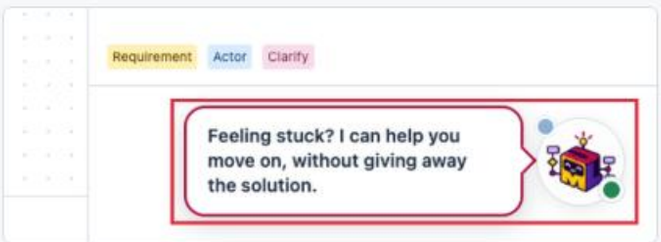
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



The video speeds up the wait. A bubble next to UMLBuddy says "Feeling stuck? I can help you move on...". The bubble leaves the workspace as it is and disappears when the student resumes editing.

▶ Watch this moment in the video (1:44)

Timely: the feedback arrives while the student's attempt is still in progress (for example, when they seem stuck or make a structural mistake), and not only when they explicitly ask, which may be too late. This keeps the feedback useful while the student is still motivated and can still correct their work.

5. Inactivity prompt

Timely

If the student has been inactive for a few minutes and the diagram is still incomplete, a bubble appears next to the tutor offering help, without opening anything on its own. The aim is to reach a student who is stuck before they give up, without interrupting one who is simply thinking.

This mechanism helps feedback arrive at the right time.

1 2 3 4 5

I don't know / not shown in the video

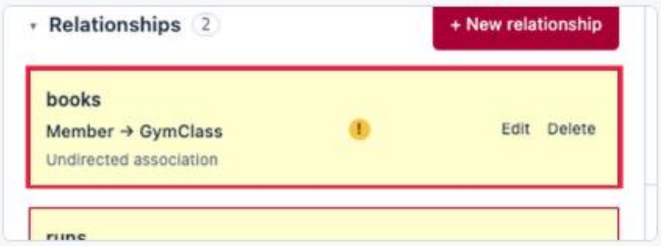
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



The "books" relationship has no multiplicity, so a warning (!) appears immediately. A program checks this independently of the language model.

▶ Watch this moment in the video (2:02)

Timely: the feedback arrives while the student's attempt is still in progress (for example, when they seem stuck or make a structural mistake), and not only when they explicitly ask, which may be too late. This keeps the feedback useful while the student is still motivated and can still correct their work.

6. Checks while editing

Timely

While the student edits, a set of rules checks for structural problems (for example, a relationship without multiplicity) and flags them immediately, without consulting the language model. The aim is to point out the mistake at the moment it is made, without waiting for the student to ask.

This mechanism helps feedback arrive at the right time.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



The "Process step" selector shows the step inferred from the student's diagram alone. The student can change it manually.

▶ Watch this moment in the video (1:30)

Contextualized: the feedback adapts to the stage of the task the student is at (for example, identifying classes or defining relationships), without imposing a fixed order between stages. This helps the student relate the hints to their own progress rather than to an abstract list of advice.

7. Detecting the current modeling step

Contextualized

From the student's diagram, the tool infers which modeling step the student is at (for example, identifying classes or defining relationships) and adapts its hints to that step, without imposing an order. The aim is for the hint to fit the moment rather than be generic advice; the student can change the step by hand.

This mechanism helps feedback fit the context.

1 2 3 4 5

I don't know / not shown in the video

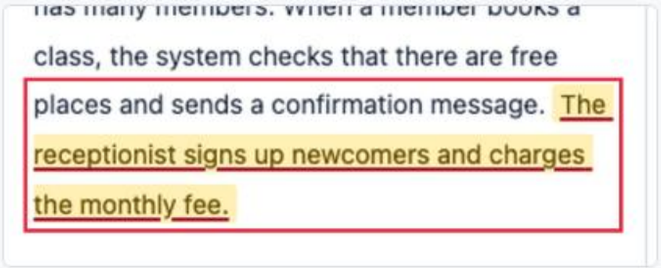
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



The phrase the tutor points to is color-coded by type (requirement, actor, or needs clarification) and underlined because it is still missing from the diagram.

▶ Watch this moment in the video (3:04)

Contextualized: the feedback adapts to the stage of the task the student is at (for example, identifying classes or defining relationships), without imposing a fixed order between stages. This helps the student relate the hints to their own progress rather than to an abstract list of advice.

8. Shared vocabulary for highlights

Contextualized

The parts of the exercise statement that the tutor mentions are color-coded by type (requirement, actor, needs clarification) and underlined if they are not yet in the diagram. The aim is for the student to see, on the statement itself, what remains to be modeled.

This mechanism helps feedback fit the context.

1 2 3 4 5

I don't know / not shown in the video

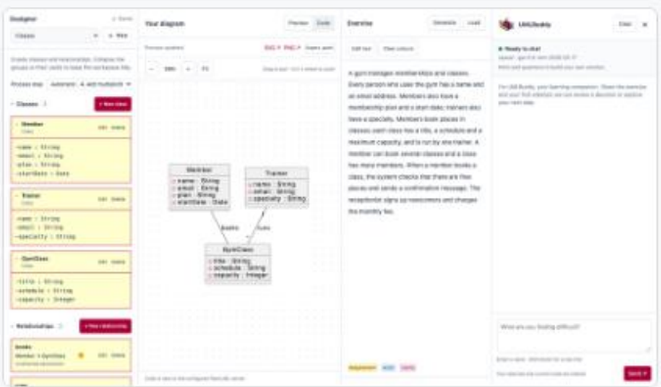
1 · Strongly disagree 5 · Strongly agree

This mechanism would get in the student's way while modeling.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree



The tutor opens next to the form, the diagram, and the exercise prompt and already has access to all three.

▶ Watch this moment in the video (2:15)

Everything in one workspace

It is useful for students to have the tutor in the design workspace so they can avoid copying the diagram and exercise prompt into a separate chat.

1 2 3 4 5

I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

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▶ Watch the video

Usefulness and adoption

Rate six statements from 1 to 5 about ModelaU's usefulness and whether you would adopt it in your teaching.

I would prefer my students to use ModelaU over a general-purpose chat for these exercises.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

Using ModelaU would help my students learn modeling.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

ModelaU's mechanisms are easy to understand.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

ModelaU fits the way I teach modeling.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

I would try ModelaU in one of my courses.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

I worry that students might rely on the tool instead of thinking for themselves.

1 2 3 4 5 I don't know / not shown in the video

1 · Strongly disagree 5 · Strongly agree

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Feedback on ModelaU

All questions are optional. Your written feedback is as helpful as a rating.

What did you like about ModelaU?

What did you dislike about ModelaU?

Any other comments

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Submit responses

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▶ Watch the video